



Security Operations Center (SOC) Fundamentals, Security Monitoring and Incident Response

1. Introduction

A Security Operations Center (SOC) is a centralized unit within an organization responsible for monitoring, detecting, analyzing, and responding to cybersecurity incidents. The SOC acts as the command center for an organization's cybersecurity operations, ensuring that systems, networks, and sensitive data remain protected from cyber threats.

SOC teams continuously monitor network traffic, system logs, and security alerts generated by various security tools. By analyzing these events in real time, analysts can detect suspicious activities such as unauthorized access attempts, malware infections, or abnormal user behavior.

The SOC also plays a crucial role in incident response, where security teams investigate security alerts, identify potential threats, contain attacks, and restore affected systems.

Through continuous monitoring and proactive threat detection, SOC teams help organizations reduce the impact of cyberattacks and maintain the security and integrity of their digital infrastructure.

2. Objectives of the Study

The main objective of this study is to understand the functioning and importance of a Security Operations Center in modern cybersecurity environments. The study focuses on learning both theoretical concepts and practical tools used by SOC analysts.

The specific objectives include:

- To understand the concept and working principles of a Security Operations Center and how it protects organizational systems from cyber threats.
- To learn the fundamentals of security monitoring, including how analysts detect suspicious activities using system logs and network data.
- To study commonly used SOC security tools, such as SIEM platforms, intrusion detection systems, vulnerability scanners, and endpoint monitoring tools.
- To understand the incident response workflow, including how cybersecurity teams detect, investigate, and respond to security incidents.
- To gain knowledge of important cybersecurity frameworks and security concepts, which help organizations establish standardized security practices.

3. SOC Fundamentals and Operations

The Security Operations Center plays a critical role in maintaining the security posture of an organization. It is responsible for detecting threats, responding to security incidents, and ensuring the protection of digital assets such as servers, databases, networks, and user devices.

SOC operations involve continuous monitoring of system activity, log analysis, threat detection, and incident response. By combining advanced security tools with skilled analysts, the SOC can identify potential cyber threats before they cause serious damage to an organization.



SOC teams follow well-defined processes and frameworks to ensure that security events are handled efficiently and consistently.

3.1 Purpose of SOC

The main purpose of a Security Operations Center is to protect an organization's information systems and networks from cyber threats.

One of the key functions of a SOC is proactive threat detection, where security analysts continuously monitor logs and network traffic to identify suspicious activities that may indicate a cyberattack.

Another important function is incident response and remediation. When a security incident is detected, the SOC team investigates the issue, identifies the source of the attack, and takes steps to contain and remove the threat.

SOC teams also perform continuous security monitoring, which ensures that systems are monitored at all times to detect potential threats as early as possible.

Additionally, SOC operations support compliance and security governance, ensuring that organizations follow cybersecurity regulations and maintain proper security controls to protect sensitive information.

3.2 SOC Roles

SOC teams are typically organized into different roles based on responsibilities and levels of expertise.

A **Tier 1 Analyst**, also known as a triage analyst, is responsible for monitoring security alerts generated by security tools such as SIEM systems. They review alerts, identify false positives, and escalate suspicious events to higher-level analysts.

A **Tier 2 Analyst** performs deeper investigation and analysis of security incidents. They examine logs, analyze attack patterns, and determine the scope and severity of the incident.

A **Tier 3 Analyst**, often called a threat hunter, is the most experienced member of the SOC team. They conduct advanced threat hunting, analyze malware, and investigate complex cyberattacks that require specialized expertise.

The **SOC Manager** oversees the entire SOC operation. They manage the security team, ensure proper incident response procedures are followed, and coordinate communication between technical teams and management.

4. Security Monitoring Basics

Security monitoring is the continuous process of observing and analyzing system activities, network traffic, and security logs to identify potential threats. It is one of the most important functions of a Security Operations Center.

Through security monitoring, organizations can detect abnormal behavior, unauthorized access attempts, and malicious activities before they lead to serious security breaches.

Security monitoring involves collecting data from multiple sources such as servers, firewalls, network devices, and applications. This data is analyzed using specialized tools to identify suspicious patterns and indicators of compromise.



4.1 Monitoring Objectives

The primary objective of security monitoring is to detect anomalies in system behavior. For example, if a user suddenly logs in from an unusual location or performs unusual actions, it may indicate a potential security issue.

Another objective is to identify unauthorized access attempts, such as repeated failed login attempts or brute-force attacks on user accounts.

Security monitoring also aims to detect potential cyber attacks, including malware infections, data exfiltration attempts, or suspicious network traffic patterns.

By detecting these activities early, organizations can take preventive actions to minimize the impact of cyber threats.

4.2 Monitoring Tools

Security monitoring relies on various tools that help collect and analyze security data.

Splunk is a widely used Security Information and Event Management (SIEM) platform that collects logs from different systems and provides powerful search and analysis capabilities.

Elastic Stack, also known as ELK Stack, is another popular platform used for log collection, processing, and visualization. It includes Elasticsearch for searching logs, Logstash for processing logs, and Kibana for data visualization.

Wireshark is a network packet analysis tool that allows analysts to capture and inspect network traffic in detail. It helps identify suspicious communication patterns and analyze potential network-based attacks.

5. Log Management Fundamentals

Log management is an essential component of cybersecurity operations. It involves collecting, storing, and analyzing logs generated by various systems and applications.

Logs contain valuable information about system activities, user actions, and security events. By analyzing logs, SOC analysts can detect suspicious behavior, identify security incidents, and investigate potential cyberattacks.

Effective log management helps organizations maintain visibility into their systems and quickly detect abnormal activities.

5.1 Log Lifecycle

The log lifecycle consists of several stages that ensure logs are properly managed and analyzed.

The first stage is Log Collection, where logs are gathered from different sources such as servers, firewalls, applications, and network devices.

The next stage is Log Normalization, where logs from different systems are converted into a standardized format so that they can be easily analyzed.

After normalization, logs are stored in a centralized storage system during the Log Storage phase.

The Log Retention phase ensures that logs are stored for a specified period according to organizational policies and regulatory requirements.

Finally, during Log Analysis, security analysts examine the logs to identify suspicious activities and potential security threats.



6. Security Tools Overview

Various security tools were studied and tested to understand how SOC teams detect and respond to cyber threats. These tools help monitor systems, identify vulnerabilities, and detect malicious activities.

6.1 Intrusion Detection System

An Intrusion Detection System (IDS) is used to monitor network traffic and detect suspicious activities that may indicate a cyberattack.

In this study, Snort IDS was used to detect malicious HTTP traffic by analyzing network packets and matching them with predefined detection rules.

Snort rules allow analysts to identify specific attack patterns and generate alerts when suspicious traffic is detected.

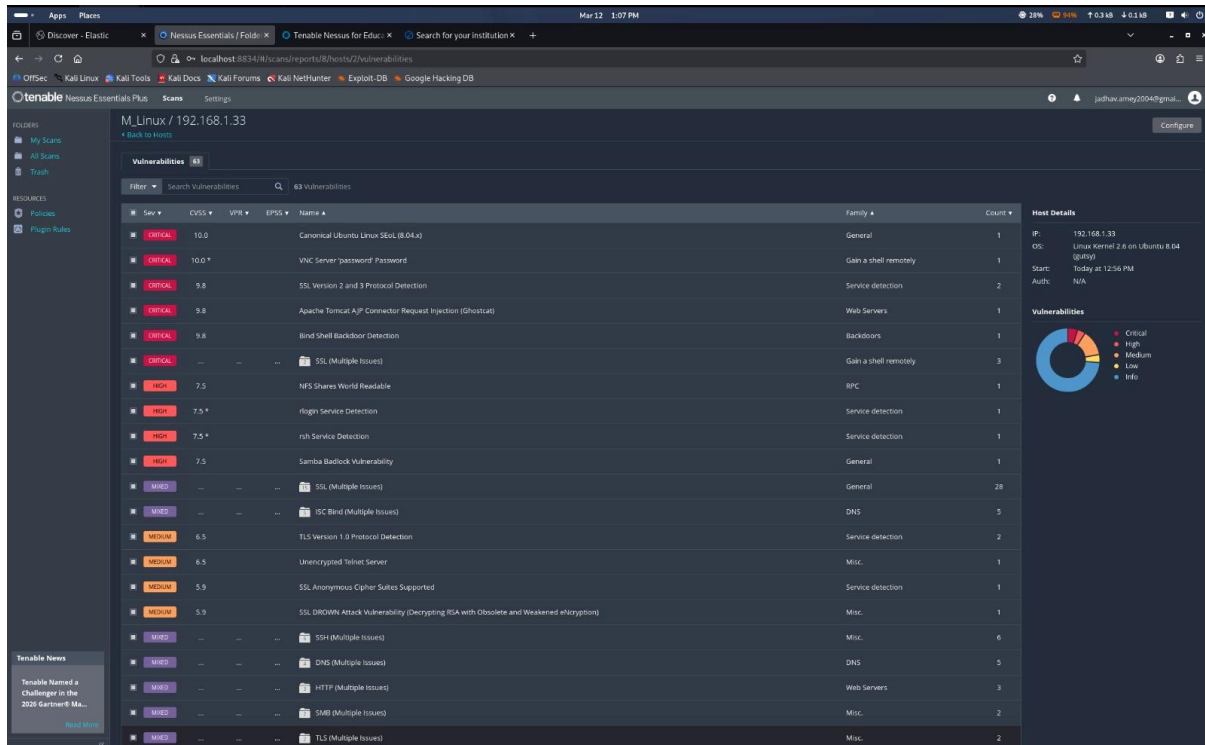
```
(kali@kali)~$ sudo nano /tmp/test.rules
(kali@kali)~$ sudo snort -R /tmp/test.rules -i lo -A alert_fast
-----
Snort++ 3.11.1.0
-----
loading rule args:
loading /tmp/test.rules:
finished /tmp/test.rules:
finished rule args:
-----
rule counts
total rules loaded: 1
text rules: 1
option chains: 1
chain headers: 1
-----
port rule counts
any      tcp      udp      icmp      ip
total    1        1        1        1
-----
fast pattern groups
any: 8
-----
search engine (ac_bnf)
instances: 4
patterns: 4
pattern chars: 56
num states: 56
num match states: 4
memory scale: KB
total memory: 5.89453
pattern memory: 0.207031
match list memory: 0.5625
transition memory: 4.625
-----
pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
# [0] lo
03/12-11:14:24.969563 [**] [1:1000001:1] "ALARM: Malicious String Found" [**] [Priority: 0] {ICMP} 127.0.0.1 -> 127.0.0.1
03/12-11:14:45.382278 [**] [1:1000001:1] "ALARM: Malicious String Found" [**] [Priority: 0] {ICMP} 127.0.0.1 -> 127.0.0.1
03/12-11:14:46.932622 [**] [1:1000001:1] "ALARM: Malicious String Found" [**] [Priority: 0] {ICMP} 127.0.0.1 -> 127.0.0.1
03/12-11:14:48.650875 [**] [1:1000001:1] "ALARM: Malicious String Found" [**] [Priority: 0] {ICMP} 127.0.0.1 -> 127.0.0.1
03/12-11:14:49.975285 [**] [1:1000001:1] "ALARM: Malicious String Found" [**] [Priority: 0] {ICMP} 127.0.0.1 -> 127.0.0.1
03/12-11:14:50.698654 [**] [1:1000001:1] "ALARM: Malicious String Found" [**] [Priority: 0] {ICMP} 127.0.0.1 -> 127.0.0.1
03/12-11:14:51.637446 [**] [1:1000001:1] "ALARM: Malicious String Found" [**] [Priority: 0] {ICMP} 127.0.0.1 -> 127.0.0.1
03/12-11:14:52.374244 [**] [1:1000001:1] "ALARM: Malicious String Found" [**] [Priority: 0] {ICMP} 127.0.0.1 -> 127.0.0.1
```

6.2 Vulnerability Scanning

Vulnerability scanners are used to identify security weaknesses in systems, applications, and networks.

In this study, Nessus was used to scan a system for vulnerabilities. The scanner analyzes the system configuration and compares it with known security vulnerabilities.

The scan results help security teams identify potential security risks and prioritize patching and remediation activities.



6.3 Endpoint Monitoring

Endpoint monitoring tools help track activities occurring on individual systems such as computers and servers.

Osquery was used to monitor running processes and system activities. It allows analysts to query system information using SQL-like commands.

This tool helps identify suspicious processes or unauthorized applications running on a system.

```

kali@kali:~$ sudo nano /tmp/test.rules
kali@kali:~$ sudo short -R /tmp/test.rules -i to -A alert_fast
p)~ Smort++ 3.11.1.0
-----
loading rule args:
loading /tmp/test.rules:
finished /tmp/test.rules:
finished rule args:
-----
rule counts
total rules loaded: 1
text rules: 1
option chains: 1
chain headers: 1
-----
port rule counts
tcp udp icmp ip
any 1 1 1 1
total 1 1 1 1
-----
fast pattern groups
any: 8
-----
search engine (ac_bfpa)
instances: 4
patterns: 4
pattern chars: 56
num states: 56
num match states: 4
memory scale: KB
total memory: 5.89453
pattern memory: 0.207031
match list memory: 0.5625
transition memory: 4.625
-----
pcap DAQ configured to passive.
commencing packet processing
Retry queue interval is: 200 ms
+ (3) to
03/12-11:14:24.969503 [**] [1:1000001:1] "ALARM: Malicious String Found" [**] [Priority: 0] [ICMP] 127.0.0.1 -> 127.0.0.1
03/12-11:14:45.382278 [**] [1:1000001:1] "ALARM: Malicious String Found" [**] [Priority: 0] [ICMP] 127.0.0.1 -> 127.0.0.1
03/12-11:14:46.932622 [**] [1:1000001:1] "ALARM: Malicious String Found" [**] [Priority: 0] [ICMP] 127.0.0.1 -> 127.0.0.1
03/12-11:14:48.058075 [**] [1:1000001:1] "ALARM: Malicious String Found" [**] [Priority: 0] [ICMP] 127.0.0.1 -> 127.0.0.1
03/12-11:14:49.975205 [**] [1:1000001:1] "ALARM: Malicious String Found" [**] [Priority: 0] [ICMP] 127.0.0.1 -> 127.0.0.1
03/12-11:14:50.698654 [**] [1:1000001:1] "ALARM: Malicious String Found" [**] [Priority: 0] [ICMP] 127.0.0.1 -> 127.0.0.1
03/12-11:14:51.637446 [**] [1:1000001:1] "ALARM: Malicious String Found" [**] [Priority: 0] [ICMP] 127.0.0.1 -> 127.0.0.1
03/12-11:14:52.374244 [**] [1:1000001:1] "ALARM: Malicious String Found" [**] [Priority: 0] [ICMP] 127.0.0.1 -> 127.0.0.1

```



7. Basic Security Concepts

Understanding fundamental cybersecurity concepts is important for SOC analysts to effectively detect and respond to threats.

7.1 CIA Triad

The CIA Triad is the foundation of information security.

Confidentiality ensures that sensitive data is only accessible to authorized individuals. Techniques such as encryption and access control are used to protect confidential information.

Integrity ensures that data remains accurate and unchanged unless modified by authorized users. Hashing and digital signatures are commonly used to maintain data integrity.

Availability ensures that systems and data remain accessible to authorized users when needed. Backup systems, redundancy, and disaster recovery plans help maintain availability.

7.2 Threat, Vulnerability, and Risk

Threat is any potential event or action that can cause harm to a system or organization, such as hackers, malware, or insider attacks.

Vulnerability is a weakness in a system that can be exploited by a threat. Examples include outdated software or misconfigured systems.

Risk is the likelihood that a threat will exploit a vulnerability and cause damage to an organization.

Understanding these concepts helps organizations identify security weaknesses and implement appropriate security controls.

7.3 Security Models

Defense in Depth is a security strategy that uses multiple layers of security controls to protect systems. If one layer fails, other layers continue to provide protection.

Zero Trust is a modern security model based on the principle of “Never Trust, Always Verify.” In this model, every user and device must be authenticated and authorized before accessing resources, regardless of their location.

8. Security Operations Workflow

The SOC follows a structured workflow to manage security incidents effectively.

The first stage is Detection, where security tools generate alerts based on suspicious activities.

Next is Triage, where analysts review alerts and prioritize them based on severity and potential impact.

During Investigation, analysts analyze logs, system data, and indicators of compromise to determine whether the alert represents a real threat.

Finally, in the Response stage, security teams take action to contain the threat, remove malicious activity, and restore affected systems.



9. Incident Response Lifecycle

Incident response is the process of managing and responding to cybersecurity incidents. The lifecycle begins with Preparation, where organizations establish policies, tools, and procedures for handling incidents.

During Identification, security teams detect and confirm that a security incident has occurred. In the Containment stage, steps are taken to prevent the threat from spreading to other systems.

During Eradication, the root cause of the incident is removed from the environment.

The Recovery stage involves restoring affected systems and ensuring they are functioning normally.

Finally, the Lessons Learned stage helps organizations analyze the incident and improve their security practices to prevent future attacks.

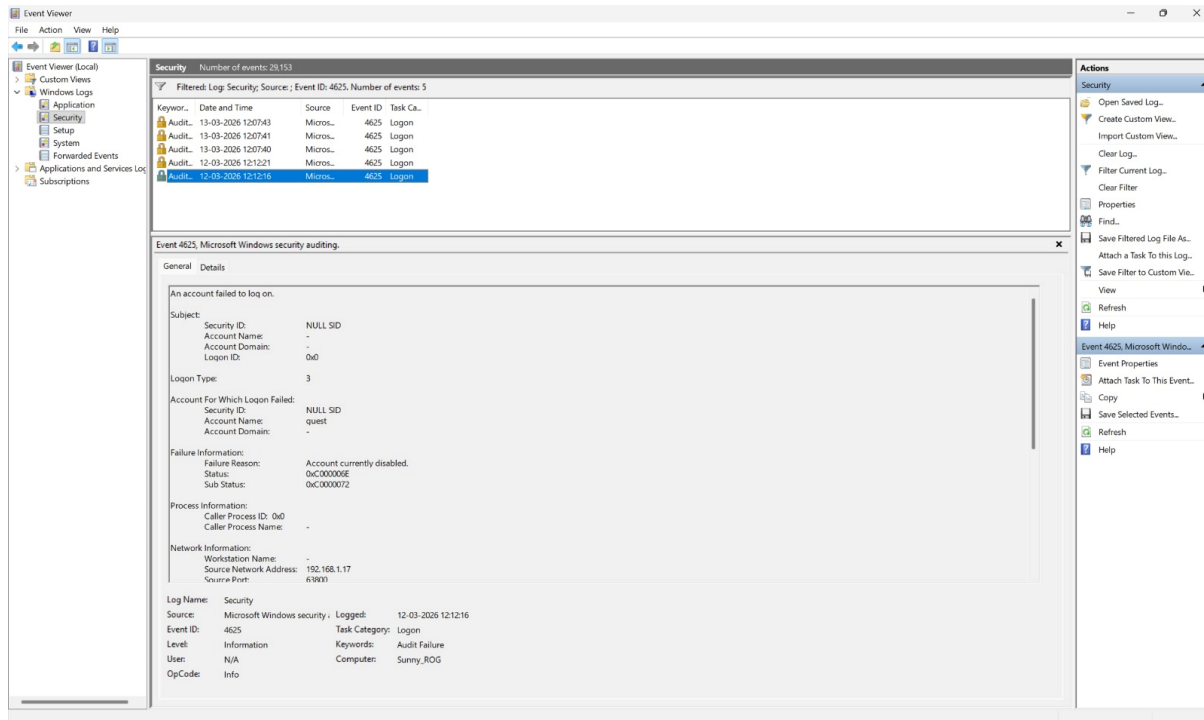
10. Log Analysis and Windows Artifact Examination

Log analysis is an essential technique in digital forensics used to detect suspicious or unauthorized activities within a system. In this task, Windows Security logs were analyzed to identify failed login attempts, and forensic tools were used to examine system artifacts that record recent user activity.

10.1 Windows Event Viewer Analysis

Windows Event Viewer is a built-in administrative tool that records system, application, and security events. These logs help investigators monitor authentication attempts, system changes, and possible security incidents.

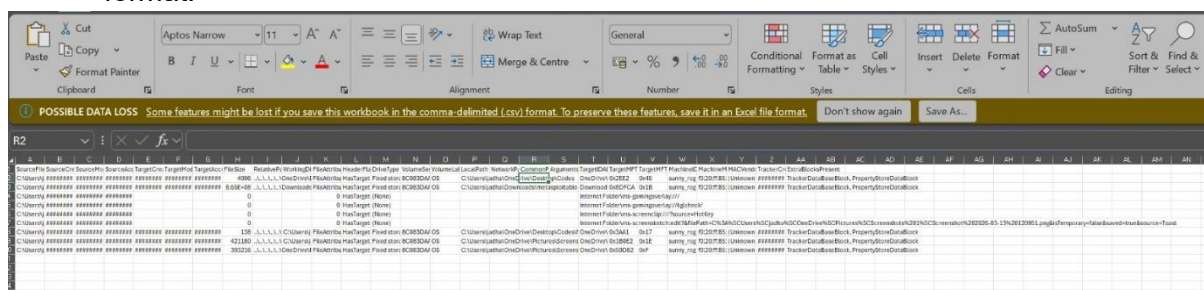
- The Security log was analyzed to identify failed authentication attempts.
- The log was filtered using Event ID 4625, which represents a failed logon event.
- Event ID 4625 is commonly used to detect:
 - Unauthorized login attempts
 - Brute-force attacks
 - Access attempts to disabled accounts
- Important fields observed in the event details include:
 - Account Name – Username used in the login attempt
 - Logon Type – Indicates the type of login (e.g., network, interactive)
 - Failure Reason – Explanation for authentication failure
 - Source Network Address – IP address of the system initiating the login attempt
- The log entries showed multiple failed login attempts for the Guest account, which was disabled, and the source IP address was recorded as 192.168.1.17.



10.2 Windows Shortcut Artifact Analysis using LECmd

To further analyze system activity, LECmd, a forensic tool developed by Eric Zimmerman, was used to parse Windows shortcut files (.lnk). These files are automatically created by Windows when users open files or applications.

- The tool was executed to analyze shortcut files located in the Windows Recent folder.
- The following command was used to parse the files and export the results to CSV format:





```
C:\Users\jadha\Downloads\LECmd>.\LECmd.exe -d "C:\Users\jadha\AppData\Roaming\Microsoft\Windows\Recent" --csv C:\Users\Public\Forensics
LECmd version 1.5.1.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/LECmd

Command line: -d C:\Users\jadha\AppData\Roaming\Microsoft\Windows\Recent --csv C:\Users\Public\Forensics

Looking for lnk files in C:\Users\jadha\AppData\Roaming\Microsoft\Windows\Recent

Found 9 files

Processing C:\Users\jadha\AppData\Roaming\Microsoft\Windows\Recent\Codes.lnk

Source file: C:\Users\jadha\AppData\Roaming\Microsoft\Windows\Recent\Codes.lnk
Source created: 2026-03-12 06:42:41
Source modified: 2026-03-12 10:40:24
Source accessed: 2026-03-13 06:48:06

--- Header ---
Target created: 2025-05-16 04:28:18
Target modified: 2025-09-03 04:37:06
Target accessed: 2026-03-12 10:40:24

File size (bytes): 4,096
Flags: HasTargetIdList, HasLinkInfo, HasRelativePath, IsUnicode, DisableKnownFolderTracking
File attributes: FileAttributeReadOnly, FileAttributeDirectory, FileAttributeArchive, FileAttributeReparsePoint
Icon index: 0
```

11. Documentation of Security Events

Documenting security events is an important practice in cybersecurity and digital forensics. It helps security analysts maintain records of suspicious activities, track incidents, and support further investigations. Proper documentation ensures that security events can be analyzed later and assists in incident response procedures.

11.1 Security Event Documentation Template

To document security events in a structured way, a template was created containing the following fields:

Field	Description
Date/Time	The exact timestamp when the event occurred
Source IP	IP address from which the event originated
Event ID	Unique identifier of the security event
Description	Brief explanation of the event



```
Since you have already generated real data in your previous labs (the **Snort** alert for 'malicious.com' and the **Event Viewer** log for failed logins), we are going to use that actual evidence to f
In a professional Security Operations Center (SOC), this is called an **Incident Log** or **Case Documentation**.
---
### **1. The Professional Security Event Template**
Copy this markdown structure into a new file (or a Note app) to use as your official template.
| Field | Description |
| --- | --- |
| **Incident ID** | Unique identifier (e.g., INC-20260313-001) |
| **Date/Time (UTC)** | Exact timestamp from the logs |
| **Severity** | Low / Medium / High / Critical |
| **Source IP** | The address where the attack originated |
| **Event ID / Type** | Windows Event ID or Snort SID |
| **Description** | A brief summary of what happened |
| **Action Taken** | Steps you took to mitigate the threat |
| **Status** | Open / Investigating / Resolved |
---
### **2. Lab Practice: Documenting Your "Real" Events**
Let's document two events based on your work today.
#### **Event A: Network Intrusion Attempt (Snort)**
Based on your successful Snort alert from earlier.
* **Date/Time:** 2026-03-12 11:20:00 IST
* **Source IP:** 127.0.0.1 (Local Simulation)
* **Event ID:** SID: 10000001
* **Description:** "Malicious Domain Detected." Outbound request containing the string 'malicious.com' was captured on interface 'lo'.
* **Action Taken:** Analyzed process logs using Osquery; identified 'curl' command as the source. No further action needed as it was a verified test.
#### **Event B: Brute-force Attempt (Windows)**
Based on the failed logins you generated.
* **Date/Time:** 2026-03-13 06:40:12 IST
* **Source IP:** 192.168.1.6 (Your Kali IP)
* **Event ID:** Windows 4625
* **Description:** "Multiple failed logins." 5 consecutive login failures for user 'hacker_test'.
* **Action Taken:** Verified account was not locked; source IP belongs to internal security testing machine. Incident marked as "False Positive - Authorized Testing."
---
### **3. Advanced Task: The "Case Study" Analysis**
To make this lab "Elite," you need to add a **Lessons Learned** section. This is what differentiates a junior analyst from a senior lead.
**Write this under your logs:**
> **Post-Incident Analysis:**
> While the Snort alert detected the network traffic, the Windows Event Log provided the "User Context" (the account name being targeted). To improve our security posture, we should implement a **Corr
---
### **4. Organizing the Evidence**
Remember the **LECmd** output you just generated? In a real report, you would attach the CSV as "Appendix A."
**Practice Step:**
Take the CSV file you created ('C:\Users\Public\Forensics\..._LECmd_Output.csv') and rename it to 'INC-20260313-001_Evidence.csv'. This keeps your forensics files organized and ready for a court of la
---
### **Summary of the Workflow**
1. **See it** (Snort/Event Viewer).
2. **Verify it** (Osquery/Forensics).
3. **Log it** (The Template).
4. **Improve it** (Lessons Learned).
**Would you like me to help you create a Python script that automatically takes a CSV file and converts it into this clean Markdown table for your reports?** This will save you hours of manual typing.
```

11.2 Importance of Security Event Documentation

Maintaining proper documentation of security events provides several benefits:

- Helps track and analyze suspicious system activities.
- Supports incident response and forensic investigations.
- Provides evidence for security audits and compliance requirements.
- Assists in identifying recurring attack patterns.

SourceFile											
A	B	C	D	E	F	G	H	I	J	K	L
SourceFile	SourceCre	SourceMoc	SourceAcc	TargetCrea	TargetMod	TargetAcce	FileSize	RelativePath	Working	FileAttributes	HeaderFlags
C:\Users\	#####	#####	#####	#####	#####	#####	4096	..\..\..\OneDrive\Desktop\Codes		FileAttributeReadonly, FileAttribu	HasTargetIdList, HasLin
C:\Users\	#####	#####	#####	#####	#####	#####	865084584	..\..\..\Downloads\metasploit-table-linux-2.0.0.zip		FileAttributeArchive	HasTargetIdList, HasLin
C:\Users\	#####	#####	#####				0			0	HasTargetIdList, IsUnico
C:\Users\	#####	#####	#####				0			0	HasTargetIdList, IsUnico
C:\Users\	#####	#####	#####				0			0	HasTargetIdList, IsUnico
C:\Users\	#####	#####	#####				0			0	HasTargetIdList, IsUnico
C:\Users\	#####	#####	#####	#####	#####	#####	138	..\..\..\OneDrive\Desktop\Codes\New Text Docume	C:\Users\	FileAttributeArchive, FileAttributeR	HasTargetIdList, HasLin
C:\Users\	#####	#####	#####	#####	#####	#####	421180	..\..\..\OneDrive\Pictures\Screenshots 1\Screensho	C:\Users\	FileAttributeArchive, FileAttributeR	HasTargetIdList, HasLin
C:\Users\	#####	#####	#####	#####	#####	#####	393216	..\..\..\OneDrive\Pictures\Screenshots 1		FileAttributeReadonly, FileAttribu	HasTargetIdList, HasLin



M	N	O	T	U	V	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI
1	DriveType	VolumeSer	VolumeLat	TargetIDat	TargetMFT	TargetMFT	MachineM	MACVend	TrackerCre	ExtraBlocksPresent							
2	Fixed stora	8CD83DAA	OS	OneDrive\	0x2EE2	0x45	f0:20:ff:85	(Unknown	#####	TrackerDataBaseBlock, PropertyStoreDataBlock							
3	Fixed stora	8CD83DAA	OS	Downloads\	0x8DFCA	0x1B	f0:20:ff:85	(Unknown	#####	TrackerDataBaseBlock, PropertyStoreDataBlock							
4	(None)			Internet Folder\	ms-gamingoverlay\	///											
5	(None)			Internet Folder\	ms-gamingoverlay\	///kgi/check/											
6	(None)			Internet Folder\	ms-screenshot\	///?source=HotKey											
7	(None)			Internet Folder\	ms-screensketch\	edit?&filePath=C%3A%5CUsers%5Cjadhya%5COneDrive%5CPictures%5CScreenshots%201%5CScreenshot%202026-03-13%20120951.p											
8	Fixed stora	8CD83DAA	OS	OneDrive\	0x3AA1	0x17	f0:20:ff:85	(Unknown	#####	TrackerDataBaseBlock, PropertyStoreDataBlock							
9	Fixed stora	8CD83DAA	OS	OneDrive\	0x1B0E2	0x1E	f0:20:ff:85	(Unknown	#####	TrackerDataBaseBlock, PropertyStoreDataBlock							
10	Fixed stora	8CD83DAA	OS	OneDrive\	0x50D82	0xF	f0:20:ff:85	(Unknown	#####	TrackerDataBaseBlock, PropertyStoreDataBlock							
11																	
12																	
13																	
14																	

12. Setting Up Security Monitoring Dashboards

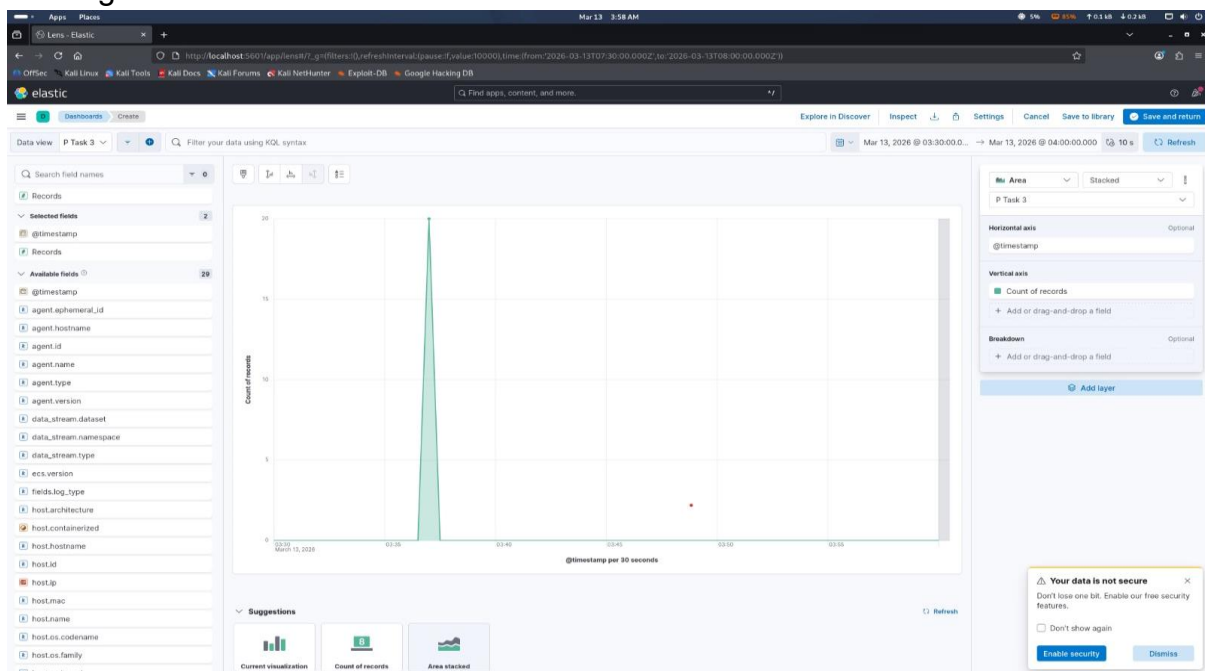
Security monitoring dashboards allow analysts to visualize system logs and detect unusual patterns in real time. Tools such as Kibana and Grafana are commonly used to monitor security events and generate visual reports.

12.1 Creating Security Visualizations

Monitoring dashboards were configured in the Elastic Stack to visualize security-related log data collected from the monitored system. Using the Elastic Lens visualization tool, log records were analyzed and displayed in graphical form to better understand system activity.

One visualization created was an **area chart showing the count of log records over time**. The timestamp field was used as the horizontal axis, while the vertical axis displayed the **number of records generated within each time interval**.

This visualization helps analysts identify patterns in log generation and detect unusual spikes in activity. A sudden increase in the number of records may indicate suspicious behavior, security alerts, or abnormal system events that require further investigation.





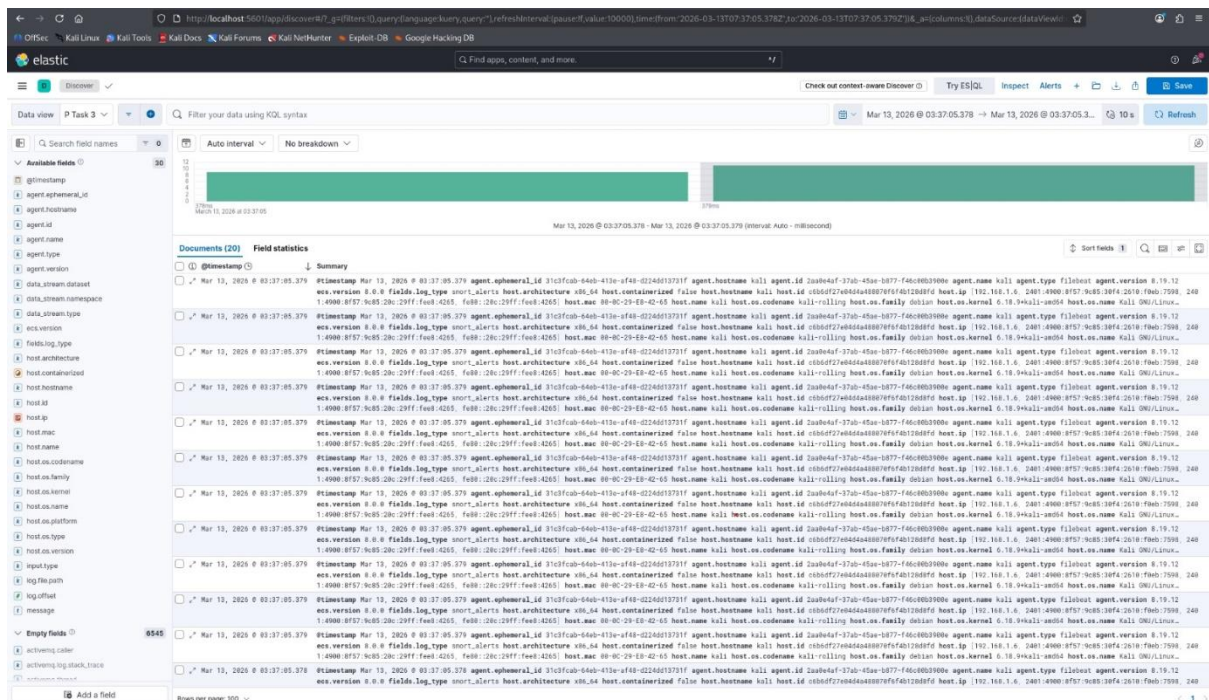
12.2 Log Analysis Using Discover

The **Elastic Discover** interface was used to examine the detailed log entries collected by the system. Logs were shipped using **Filebeat**, which forwards security-related log data to Elasticsearch for indexing and analysis.

The Discover page displays various log fields including:

- timestamp – indicates the exact time the event occurred.
- agent.hostname – identifies the system generating the logs.
- host.ip – shows the IP address of the monitored host.
- agent.type – indicates the log shipper
- fields.log_type – specifies the type of security log being recorded (Snort alerts).

These logs provide detailed information about system and network activities. Analysts can filter, search, and analyze the logs to identify potential security incidents or abnormal behaviors.



12.3 Benefits of Security Monitoring Dashboards

Security monitoring dashboards provide several important benefits:

- **Real-time monitoring** of system and security logs.
- **Quick identification of unusual log activity or spikes.**
- **Improved visibility of security events across the system.**
- **Efficient investigation of logs using visualization and search tools.**

By using dashboards and log analysis tools in the Elastic Stack, security analysts can monitor system activity more effectively and respond quickly to potential security threats.



13. Configure Alert Rules

Alert rules were configured in the Elastic Security platform to automatically detect suspicious activities and generate alerts. Detection rules help security analysts monitor system behavior and identify potential threats without manual log inspection.

A detection rule was created and executed to monitor security events in the environment. The rule checks log data stored in Elasticsearch and triggers alerts when the defined conditions are met.

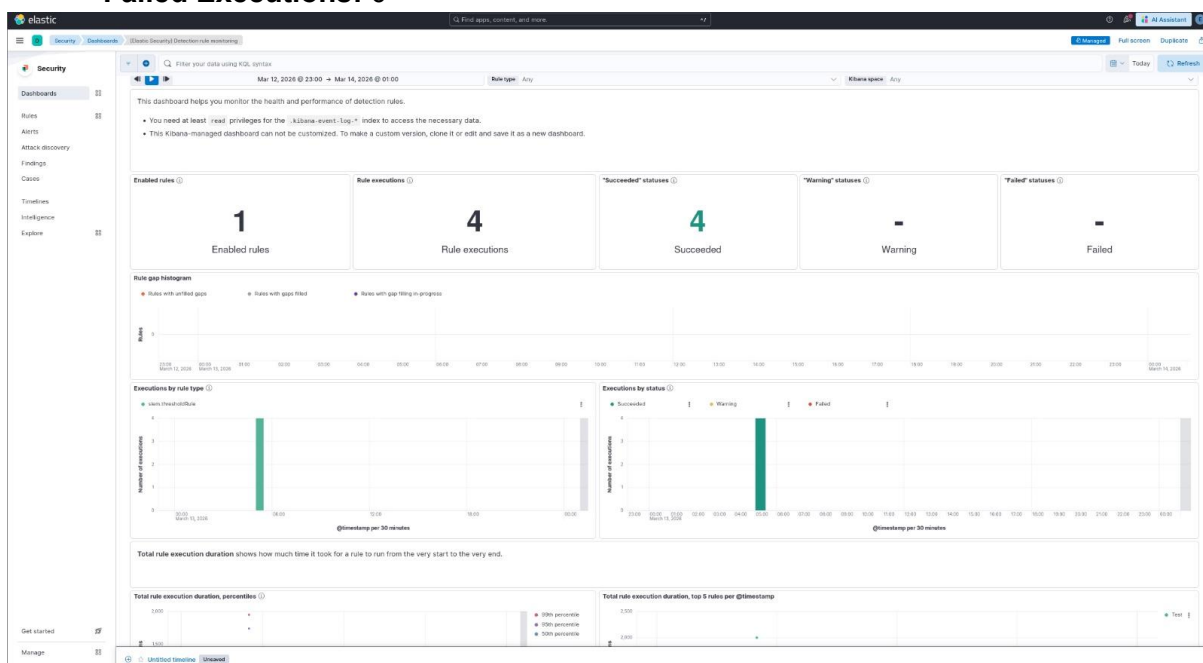
Detection Rule Configuration

- **Rule Name:** Test Detection Rule
- **Rule Type:** Threshold Rule
- **Condition:** Detect multiple security-related log events within a defined time interval
- **Data Source:** Logs collected through Filebeat and indexed in Elasticsearch
- **Execution Interval:** Runs periodically to analyze incoming log data

After configuring the rule, it was executed to test its functionality. The dashboard shows the rule monitoring results including the number of rule executions and their status.

From the dashboard results:

- **Enabled Rules:** 1
- **Rule Executions:** 4
- **Successful Executions:** 4
- **Warnings:** 0
- **Failed Executions:** 0



Alert Validation

The configured rule was validated by generating test log activity and verifying that the rule executed correctly in the Elastic Security dashboard. The monitoring panel confirmed that the rule ran successfully and processed the log data.

This validation ensures that the detection rule can be used to automatically monitor logs and detect suspicious events in the system.